Data and Applications

Requirement Analysis for Hashira Market

[Project Phase I]

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1.1 idea formulation

Hashira Market is a fantasy financial ecosystem set in the Demon Slayer universe, where heroes, demons, and villains act as investors in a supernatural stock exchange. Traditional Demon Slayer roles are reimagined as financial participants. Hashira and Slayers invest in companies producing demon-slaying gear, training methods, and potions, while demons fund dark arts firms and blood demon art labs. The database mirrors real-world stock market concepts like dividends, revenue, profits, and promoters but here, the traders are warriors and demons.

In this world, market dynamics revolve around fights, spells, and company performance. Companies in sectors such as Weapons, Potions, Techniques, Armor, and Tools go public through IPOs, and their stock prices fluctuate based on fight outcomes and magical influences. A hero's victory can boost stocks for slayer equipment while causing a drop in demon-related investments. Each investor maintains a personal portfolio, tracks their win rate in battles, mentors other characters, and earns dividends from profitable ventures proving that, in Hashira Market, both strength and strategy determine financial success.

1.2 User Identification & Interaction

- Promoters(Founders and market boosters).

Interactions: Create companies and IPOs, post promotional news (NEWS_EFFECT), update firm details (CEO, MarketCap), monitor promoted portfolios, generate engagement reports, and manage promotion tenure.

- Investors (Core traders and fighters)

Interactions: Register profiles (INVESTOR, HASHIRA/UPPER_RANK_DEMON), execute trades (TRANSACTION, PORTFOLIO), join battles (FIGHT, FIGHT_SCORE), cast spells (MAGIC_SPELL), track P&L and stock values, earn dividends, and participate in mentorships.

- Guild Analysts(Market intelligence experts)

Interactions: Generate sector and performance reports (SECTOR–COMPANY–STOCK), analyze battle impacts (FIGHT–NEWS_EFFECT–STOCK), assess risks, study investor sentiment, and track spell-based manipulations.

- Market Administrator(System regulators.)

Interactions: Approve battles, enforce integrity rules, handle deletions and audits, monitor fraud, manage skill and spell limits, enforce compliance, and maintain investor and company records.

1.3 Purpose of the Database

Importance of the Database in Hashira Market

The database is critical for maintaining consistency and accuracy across interconnected market events, where a single battle can simultaneously affect multiple stocks, investor portfolios, and performance statistics. Unlike file-based solutions, the database ensures:

- Atomic Transactions: Updates from battles—including outcomes, news generation, stock price adjustments, and portfolio recalculations—either all succeed or fail together, preventing inconsistent states.
- Referential Integrity: Cascading rules automatically remove dependent data (e.g., BattleScores, NewsEffects) when a battle is deleted, avoiding orphaned or corrupted records.
- Concurrent Access: Multiple investors can trade at the same time without conflicts or lost updates, which flat files cannot handle reliably.
- Complex Queries: Multi-table joins and aggregations enable detailed analysis of dividends, sector performance, portfolio P&L, and battle impacts efficiently.
- Derived Values: Views like STOCK_CURRENT_VALUE dynamically calculate real-time stock prices from BaseValue, PowerBoosts, and NewsImpacts, eliminating redundant or stale data.
    - Declarative Constraints: Automatic enforcement of data domains, uniqueness, and foreign keys reduces errors and bypassable validations.

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2 Database Requirements

Demon Slayer Stock Market System

Strong Entities

1. SECTOR: Industry categories and classifications for companies with associated risk levels (Weapons, Potions, Techniques, etc.).

- Attributes:
    - SectorID (PK, VARCHAR(10))
    - Name (VARCHAR(50), NOT NULL)
    - Description (TEXT)
    - RiskLevel (INT, CHECK BETWEEN 1 AND 10)
    - Created_Date (DATE, DEFAULT CURRENT_DATE)

2. INVESTOR: Base information common to all market participants including identity, available capital, Fight history, and performance statistics.

- Attributes:
    - InvestorID (PK, VARCHAR(10))
    - FirstName (VARCHAR(30), NOT NULL)
    - LastName (VARCHAR(30), NOT NULL)
    - Type (VARCHAR(10), CHECK IN 'Demon slayer','Demon', NOT NULL)
    - MentorID (FK, VARCHAR(10))
    - Wallet (DECIMAL(15,2), DEFAULT 0.00)
    - JoinDate (DATE, DEFAULT CURRENT_DATE)
    - TotalBattles (INT, DEFAULT 0)
    - WinRate (DECIMAL(5,2), DEFAULT 0.00)

HASHIRA and UPPER RANK are SUBCLASSES of INVESTORS

3. HASHIRA : Demon Slayer Corps-specific attributes including breathing style (Sun, Water, Flame), hierarchical rank (Hashira, Kinoe), and corps affiliation.

- Attributes:
    - InvestorID (PK/FK, VARCHAR(10))
    - BreathingTechnique (VARCHAR(50))
    - RankLevel (INT, CHECK BETWEEN 1 AND 10)
    - Corps (VARCHAR(30))

4. UPPER RANK DEMON: Demon-specific powers (Blood Demon Art abilities), position in Muzan's hierarchy, and Upper/Lower Moon ranking (1-12).

- Attributes:

- InvestorID (PK/FK, VARCHAR(10))
- BloodArt (VARCHAR(50))
- Ranklevel (INT, CHECK BETWEEN 1 AND 10)
- corps(VARCHAR(30))

5. COMPANY: Organizations that manufacture and sell magical products, including leadership, founding date, sector classification, and total market valuation.

- Attributes:
  - CompanyID (PK, VARCHAR(10))
  - Name (VARCHAR(100), NOT NULL)
  - CEO (VARCHAR(50))
  - Founded (DATE)
  - Headquarters (VARCHAR(100))
  - MarketCap (DECIMAL(18,2), DEFAULT 0.00)
  - SharesOutstanding (BIGINT, DEFAULT 1000000)

6. PRODUCT: Magical items produced by companies including Nichirin swords, healing potions, breathing techniques, with power ratings and pricing information.

- Attributes:
  - ProductID (PK, VARCHAR(10))
  - Name (VARCHAR(100), NOT NULL)
  - Type (VARCHAR(20), CHECK IN 'Weapon','Potion','Technique','Armor','Tool')
  - PowerLevel (INT, CHECK BETWEEN 1 AND 100)
  - CompanyID (FK, VARCHAR(10))
  - Price (DECIMAL(10,2))
  - ReleaseDate (DATE)

7. STOCK: Tradeable securities representing company ownership with ticker symbols, base prices, dividend payments, IPO dates, and total shares available.

- Attributes:
  - StockID (PK, VARCHAR(10))
  - Symbol (VARCHAR(10), UNIQUE, NOT NULL)

- Name (VARCHAR(100), NOT NULL)
- BaseValue (DECIMAL(10,2), NOT NULL)
- CompanyID (FK, VARCHAR(10))
- IPODate (DATE)
- Share Value

8. FIGHT: Combat events between investors with outcomes that trigger market-moving news, including location, type, duration, and financial stakes.

- Attributes:
    - FIGHTID (PK, VARCHAR(10))
    - Location (VARCHAR(100))
    - WinnerInvestorID (FK, VARCHAR(10))
    - LoserInvestorID (FK, VARCHAR(10))
    - SKILLSUSED(TEXT)
    - PRODUCTS USED(TEXT)

11. MAGIC_SPELL: Market manipulation abilities that investors can cast on stocks, including effect type (price pump/crash), power cost, duration, and cooldown restrictions.

- Attributes:
    - SpellID (PK, VARCHAR(10))
    - Invesor ID(FK,VARCHAR(50))
    - Name (VARCHAR(100), NOT NULL)
    - EffectType (VARCHAR(20), CHECK IN 'Pump','Crash','Freeze','Volatility')
    - Duration (INT)
    - CooldownPeriod (INT)
    - Description (TEXT)

12. NEWS_EFFECT: Market-moving announcements triggered by battle outcomes, including narrative description, winner type, average stock impact percentage, and severity level.

- Attributes:
    - NewsID (PK, VARCHAR(10))

- FIGHTID (FK, VARCHAR(10))
- WinnerType (VARCHAR(10), CHECK IN 'Hero','Villain','Demon')
- Description (TEXT)
- StockEffect (DECIMAL(5,2))
- ImpactLevel (VARCHAR(10), CHECK IN 'Low','Medium','High','Critical')
- Timestamp (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

13.TRANSACTION: Records all buy/sell transactions of stocks between investors, including quantity, price, date, type of transaction, and any associated fees or notes.

- Attributes:
  - TransactionID (PK, VARCHAR(15))
  - InvestorID (FK, VARCHAR(10))
  - StockID (FK, VARCHAR(10))
  - TransactionType (VARCHAR(10), CHECK IN 'Buy','Sell')
  - Quantity (INT, CHECK > 0)
  - PricePerShare (DECIMAL(10,2)) — Price at which each share was traded
  - TransactionDate (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)
  - TotalValue (DECIMAL(18,2))

Weak Entities

1. INVESTOR_SKILLS : Multiple skills per investor (combat techniques, trading abilities, analytical skills) with proficiency levels from 1-10.

- Attributes:
  - InvestorID (PK/FK, VARCHAR(10))
  - Skill (PK, VARCHAR(50))
  - SkillLevel (INT, CHECK BETWEEN 1 AND 10)

13. FIGHT_SCORE: Individual performance metrics for each investor participating in a battle, including numerical score (0-100), qualitative rating, and monetary rewards earned.

- Attributes:
  - FIGHTID (PK/FK, VARCHAR(10))
  - InvestorID (PK/FK, VARCHAR(10))

- Score (INT, CHECK BETWEEN 0 AND 100)
- Performance (VARCHAR(10), CHECK IN 'Excellent','Good','Average','Poor')
- Rewards (DECIMAL(10,2), DEFAULT 0.00)
- EffectDescription (TEXT)

Relationships

BINARY RELATIONSHIPS

1. PORTFOLIO (Investor–Stock Ownership)

- Degree: Binary
- Entities: INVESTOR, STOCK
- Cardinality: INVESTOR (1,N) : STOCK (0,N)
- Min–Max Constraint: INVESTOR (1,N) : STOCK (0,N)
- Explanation: Many-to-many ownership relationship tracking stock holdings. Each investor can hold multiple stocks with quantity, purchase price, and acquisition date.

2. PRODUCES (Company–Product Production)

- Degree: Binary
- Entities: COMPANY, PRODUCT
- Cardinality: COMPANY (1,N) : PRODUCT (1,1)
- Min–Max Constraint: COMPANY (1,N) : PRODUCT (1,1)
- Explanation: One-to-many relationship where companies manufacture multiple magical products. Each product belongs to exactly one company.

3. MENTORSHIP (Investor–Investor)

- Degree: Binary (Recursive)
- Entities: INVESTOR (Mentor, Mentee)
- Cardinality: INVESTOR (0,1) : INVESTOR (0,N)
- Min–Max Constraint: Mentee (0,1) : Mentor (0,N)
- Explanation: Hierarchical learning structure where seasoned investors guide new participants. Each mentee has at most one mentor, while mentors can guide multiple mentees.

4. PROMOTES (Company–Investor Promoters)

- Degree: Binary
- Entities: COMPANY, INVESTOR
- Cardinality: COMPANY (1,1) : INVESTOR (0,N)
- Min–Max Constraint: COMPANY (1,1) : INVESTOR (0,N)
- Explanation: Investors act as company promoters or major stakeholders. Each company has at least one promoter, while investors can promote multiple companies..

5. TRANSACTS (Investor–Stock)

- Degree: Binary
- Entities: INVESTOR, STOCK
- Cardinality: INVESTOR (1,N) : STOCK (1,N)
- Min–Max Constraint: INVESTOR (1,N) : STOCK (0,N)
- Explanation: Many-to-many capturing stock trading activity. Tracks historical prices, dividend eligibility, and market transactions.

TERNARY RELATIONSHIPs

10. CASTS_SPELL_ON (Investor–Spell–Stock)

- Degree: Ternary
- Entities: INVESTOR, SPELL, STOCK
- Cardinality: INVESTOR (0,N) : SPELL (0,M) : STOCK (0,P)
- Min–Max Constraint: INVESTOR (0,N) : SPELL (0,M) : STOCK (0,P)
- Explanation: Tracks which investor cast which spell on which stock. Includes casting timestamp, effect duration, and power consumption for each spell cast.

QUATERNARY RELATIONSHIPS

11. INVESTOR_PRODUCT_FIGHT_REWARD

- Degree: Quaternary
- Entities: INVESTOR, PRODUCT, FIGHT, STOCK
- Cardinality: 1:N:M:P
- Min–Max Constraint: INVESTOR (0,N) : PRODUCT (0,M) : FIGHT (0,P) : STOCK (0,Q)

- Explanation: Captures when an investor uses a product in a fight affecting stock value. Tracks rewards earned and effect descriptions from combat product usage.

12. INVESTOR_FIGHT_SKILL_SCORE

- Degree: Quaternary
- Entities: INVESTOR, FIGHT, INVESTOR_SKILLS, FIGHT_SCORE
- Cardinality: 1:N:M:P
- Min–Max Constraint: INVESTOR (0,N) : FIGHT (0,M) : INVESTOR_SKILLS (0,P) : FIGHT_SCORE (0,Q)
- Explanation: Connects investor's specific skills to their performance in fights. Tracks how particular skills contribute points and performance ratings in battles.

Composite attributes

- InvestorName (conceptual composite) from FirstName and LastName.
- CompanyAddress as a composite of street, headquarters location, and postal code (if explicitly modeled).

Derived attributes

- StockCurrentValue derived from BaseValue, PowerBoosts, and NewsImpacts.
- Investor's WinRate derived from number of battles won and total battles fought

Multivalued attributes

- Breathing Techniques(a hashira can know about multiple breathing techniques)
- Product type(a company can have multiple product types like armour,potions,spells etc..)

3.1 Retrieval Operations

Selection

• **Battles with specific outcomes**: Retrieve all battles where the winner was a Demon.
• **Investors in specific companies**: List the names of investors who invested in *FlameBreath Inc.*
• **News based on entity links**: Fetch all news items linked to a given battle or company.
• **Investors with certain achievements**: Retrieve investors who have earned rewards above a specific threshold.

Projection

• **Battle summary data**: Show only the Battle IDs and corresponding winners of all battles.
• **Investor reward records**: Retrieve Investor IDs and their reward points from all battles.

• **Stock performance overview**: Display only Stock IDs and their current market values.
• **Company-product catalog**: Retrieve Company IDs with their respective Product names.

Aggregators
• **Market effect totals**: Calculate the total stock impact percentage from all news effects related to Demon victories.
• **Reward averages**: Find the average reward points earned by investors in battles won by heroes.
• **Participation count**: Determine how many battles each investor has participated in.
• **Stock influence analysis**: Compute the average impact percentage of battles on different stocks.

Search
• **News phrase search**: Retrieve all news effects containing the phrase *"Stock crash"* in their description.
• **Market sentiment detection**: Find news effects mentioning *"Market boom"* or *"Market crash."*
• **Product lookup**: Search for products whose names include *"Nichirin"* or *"Potion."*
• **Investor keyword search**: Find investors whose bios mention *"Flame," "Thunder,"* or *"Beast."*

3.2 Analysis Report

Impact of Battle Outcomes on Stock Prices across Companies

"Analyze how battles won by different winner types affect the average stock price changes of associated companies, showing which type of battle leads to significant market shifts."

Correlation Between Investor Rewards and Battle Participation Frequency

"Report total and average reward points earned by investors grouped by how many battles they participated in, to identify if frequent participants benefit more."

Portfolio diversification vs win rate analysis

"Analyse if investors with diversified portfolios have better battle win rates than those with concentrated holdings"

3.3 Modification Operations

INSERT

Operations:

- Insert new **Investors** (under Demons or Hashira Mentors).
- Insert new **Companies** and their associated **Stocks**.

- Insert new **Fights** that auto-generate **News Effects** and **Stock Impact** data.
- Insert new **Portfolio** entries when investors purchase new stocks.

Valid Example:
Insert into FIGHT
<F0007, 2025/10/10, Mount Fujikasane, Hero_102, Demon_209, “Flame vs Shadow”, Winner = Hero_102>
→ Automatically generates:

- NEWS_EFFECT with +5% impact on Weapon-sector stocks.
- STOCK_AFFECTED entries linked to all affected companies.

(All integrity constraints such as valid Investor IDs, existing Stock IDs, and non-duplicate Fight IDs are auto-checked before insertion.)

UPDATE

Operations:

- Update **Investor Type** (Hero ↔ Demon) and automatically update promoter roles, mentorships, and spell permissions.
- Update **Company MarketCap** or **Stock Value** after major fights or economic events.
- Update **Portfolio Quantity** and **Wallet Balance** after stock transactions.
- Update **Fight outcomes** (e.g., in case of error correction) to recalculate related News and Stock Effects.

Valid Example:
Update Investor Type
Update INVESTOR set Type = 'Demon' where InvestorID = 'Hero_102'
→ Cascading effects:

- All mentorships under this investor are revalidated.
- Hero-only product access revoked.
- Related companies promoted by this investor flagged for review.

DELETE

Operations:

- Delete **Companies** that are delisted or merged, along with dependent Products and Stocks (soft delete preferred).

- Delete **Fight** records that were invalid (training or simulation data) — ensures related News Effects and Stock Impacts are also removed.
- Delete **Investor** data upon permanent account deactivation or disappearance from the market.
- Delete **Portfolio** or **Promotion** records when relationships end or expire.

Valid Example:
Delete from COMPANY
where CompanyID = 'C0045'
→ Cascading effects:

- Related STOCK, PRODUCT, and PROMOTES entries are archived or soft-deleted.
- All PORTFOLIO records holding those stocks are updated (set status = ‘Closed’).
- Investor notifications sent for delisted assets.